

REGEEG: A Regression-Based EEG Signal Processing in Emotion Recognition

Oscar Almanza-Conejo , Juan Gabriel Avina-Cervantes ,
Arturo Garcia-Perez , *Senior Member, IEEE*, and Mario Alberto Ibarra-Manzano , *Member, IEEE*

I. INTRODUCTION

Abstract—Electroencephalograms provide a non-invasive and effective method for studying emotion recognition and developing Artificial Intelligence (AI) models to understand human behavior and decision-making processes. This study involved testing several machine learning classification kernels to develop an accurate emotion recognition model capable of classifying emotions stimuli such as “Boring,” “Calm,” “Happy,” and “Fear” during gameplay. An emotion classifier was assessed using the publicly available database for an emotion recognition system based on EEG signals and various computer games (GAMEEMO). The signal processing method, referred to as Regression EEG (REGEEG), involves an efficient electrode pairing selector developed for EEG signal processing using a regression algorithm, rotation matrices, director vectors, and robust statistical and polynomial feature extraction. REGEEG and feature extraction methods were evaluated with 28 machine learning kernels, resulting in five kernels with classification performance above 80%, with the K-Nearest Neighbors (k-NN) based model outperforming the rest (achieving over 95% accuracy, F1-Score, and kappa-score). REGEEG performance was further validated using 30 Cross-Validation (CV) folders and 28 in the Leave-one Subject-out (LoSo) technique without impacting the average classification performance. The classification highlights revealed low variance in the CV, while the LoSo approach helped identify outliers in the GAMEEMO dataset. Furthermore, the EEG pair channels selector demonstrates superior performance in classification, indicating a correlation between features and each processed pair of channels.

Index Terms—Emotion Recognition, data regression, electroencephalogram, EEG signal processing, K-Nearest neighbors.

Received 5 March 2024; revised 8 August 2024; accepted 13 February 2025. Date of publication 10 March 2025; date of current version 4 July 2025. This work was supported in part by the Mexican National Council of Humanities, Science, and Technology (CONAHcyT) under Grant CVU 1007303, in part by the the University of Guanajuato, in part by Secretaría de Ciencia, Humanidades, Tecnología e Innovación (Secihti) under Grant #815485, and in part by the University of Guanajuato Convocatoria Institucional de Investigación Científica (CIIC), through UG Project 243/2024. (*Corresponding author: Oscar Almanza-Conejo.*)

Oscar Almanza-Conejo and Mario Alberto Ibarra-Manzano are with the Electronics Engineering Department, University of Guanajuato in the Digital Signal Processing Laboratory, Salamanca 36787, Mexico (e-mail: o.almanzaconejo@ugto.mx; ibarram@ugto.mx).

Juan Gabriel Avina-Cervantes and Arturo Garcia-Perez are with the Electronics Engineering Department, University of Guanajuato, Salamanca 36787, Mexico (e-mail: avina@ugto.mx; arturo@ugto.mx).

Digital Object Identifier 10.1109/JBHI.2025.3543729

DISCUSSING feelings and emotions is often subjective due to complex human perceptions, sometimes influenced by personal experiences. In a formal context, there are two primary Emotion Recognition (ER) model frameworks: the primary and primitive. The former relates to a theory about primary emotions [1], which suggests that happiness, anger, sadness, surprise, disgust, and fear develop in the first six months of life and are culturally universal. The latter involves self-assessment, where individuals rank the degrees of valence and arousal stimuli [2].

In Artificial Intelligence (AI), ER models commonly use electroencephalogram (EEG) signals as data to record information from both the central and peripheral nervous systems to perform specific recognition tasks. This approach is advantageous due to its relatively low cost, portability, non-invasive nature, and absence of radiation exposure [3]. Additionally, facial expression [4] and voice recognition [5], [6] are popular ER methods that are broadly studied in the context of superficial behavior.

Now, most efficient ER methods use Deep Learning (DL) architectures. For instance, [7] developed a primitive emotion recognition model using spectral analysis. They used topographic images in medical approaches to differentiate between various healthy human conditions. Contrarily, [8] studied the ER performance by thresholding primitive emotions to yield primary ones, using the power spectrum density and differential entropy features to train a deep forest classification algorithm. In close connection, [9] employed a time-frequency decomposition followed by an image classification based on primitive emotions. This study concluded that the best performance utilized two Effective Channels (EC), T8 and T7, in multiple classification algorithms.

Alternatively, [10] applied an intrinsic mode decomposition algorithm for highly discriminant multidimensional features from EEGs. The study concluded that the first decomposition level performs best on primitive classification tasks. Besides primitive emotion approaches, Positive, Negative, and Neutral (PNN) ERs have also been proposed. According to [11], the PNN emotion classifier using a Modified Differential Entropy Based on Fourth Order Spectral Moment (MD-DE) performed best in subject-dependent classification. In a closely related work, [12] used a Dynamic Graph Convolutional Neural Network (DGCNN) to analyze the spectral tensor performance of EEG localization images in a Positive and Negative ER model. Findings suggested a highly reduced spectral linkage compared with methods in the literature. Likewise, [13] used

a robust classification method by combining four different EC sets and five different EEG frequency decompositions. In brief, classification showed higher performance in a 20-EC and for δ , θ , α , and β , frequency decompositions. However, γ decomposition performed best in the 12-EC model. Moreover, using a combination of EEG-frequency decomposition features, classification obtained the highest predicted performance in a Support Vector Machines (SVM) based model.

In addition, several methodologies based on channel selection, like the previous one, have reported similar results using a reduction set of EC compared to all the EEG channels available in the database. [14] employed an ML classification to find the optimal channel selection subset in a 62 EEG channel set, suggesting a set of 10 EC, 9/10 located at the frontal lobe and a single one in the occipital lobe. Likewise, [15] reported an accurate ER model using the FC1 and CP4 channels (in a 10–20 system), highly related to the amygdala's behavior reflection in FCZ and CP1 EEGs. ML testing included the performance for a set of a single, two, four, and so on until the 62 EC performance.

In the same order of ideas, [16] completed a five primary emotions classification by computing 18 EC using brain functional connectivity patterns per primary emotion, proposing an accurate multimodal approach. Alternatively, [17] developed a robust semi-supervised classification method by evaluating the clustering performance for $S^\tau = \{C_\tau | \forall \tau = \{1, 3, 5, 7, 10, 25\}\}$ clusters and using primary and primitive datasets. Clustering showed a significant performance by increasing the labels in the primary emotions dataset; meanwhile, the primitive dataset showed a lesser sensitive performance by increasing the number of labels per cluster.

In close connection, spectral analysis noise reduction methodologies have been proposed as EEG preprocessing techniques to increase AI performance. In this respect, [18] presented a Butterworth filter in a bandwidth of [0.5, 64] Hz with a notch filter at 50Hz for A/C suppression. A more profound summary of these filtering methods is presented by [19]. Interestingly, an Empirical Mode Decomposition (EMD), based on lower frequency oscillation signals computing per decomposition level, was implemented by [20]. Here, the EMD yields five frequency decompositions, where δ , θ , α , β , and γ bands are computed, and EMD only required AF7, AF8, TP9, and TP10 EC to be completed, yielding a set of 20 signals per sample in the dataset. Long experimentation results reached classification performance of over 85% accuracy using the set of EMD signals to feed an LSTM network.

Substantially, [21] mentioned the contribution of wavelet transform and decomposition of EEGs. Consequently, [22] employed a wavelet decomposition to reach the δ , θ , α , β , and γ sub-frequencies, showing a vital noise reduction, using them to feed a feature extraction technique. Moreover, [23] conducted efficient research from 2015 to 2020, quantifying several spectral analysis techniques for feature extraction, yielding the Fourier and wavelet transforms as the top ones.

Recently, extensive experimentation has been proposed using the Database for an emotion recognition system based on EEG signals and various computer games (GAMEEMO) [24], where four computer video games are used as emotion stimuli: boring

(Low Arousal–Negative Valence, LANV), calm (Low Arousal–Positive Valence, LAPV), horror (High Arousal–Negative Valence, HANV), and funny (High Arousal–Positive Valence, HAPV) labels. In this respect, [25] presented an EEG channel selection per lobes area. Here, a lobes-dependent classification concluded in the anterior frontal area as the lower classification; meanwhile, the frontal lobe presented the top accuracy rate.

Closely related, [26] created a novel ER approach, where a 25-windowing vector per EEG is converted to a 5×5 matrix to generate four different fractal directions to compute a binary 8-bit blockchain, like Local Binary Pattern (LBP). Next, the decimal value from the blockchain increments the index in a 0–255 histogram, yielding a histogram per fractal direction. Likewise, [27] employed the continuous wavelet transform to extract scalogram images from EEGs to feed a CNN classification model with the GAMEEMO dataset. The method is like the [28] framework, where the scalogram images are extracted and sent as input to a GoogLeNet architecture, also known as Inception-v1.

In this regard, this paper proposes a mathematical algorithm to modify EEG input data behavior to apply a robust feature extraction technique. Hence, the GAMEEMO dataset is systematically used as input to compound a pair channel selection to map the FC1 and CZ channels, where, according to [29], amygdala and hippocampus signal response is associated. A 24-pair of EEG channels is also performed to compute an efficient regression algorithm as signal preprocessing for noise reduction. Besides, statistical and polynomial feature extraction is implemented in each preprocessed EEG, where the fundamental key of this algorithm is supported by a Regression on the EEGs (REGEEG). Finally, robust machine learning training and validation models are conducted to develop an accurate ER model.

The main highlights are summarized as follows:

- An efficient noise reduction method is developed using the REGEEG approach.
- Only 5 of 25 ML classification kernels obtained results superior to 80% accuracy for a 30-folder cross-validation task.
- Subject-dependent classification concluded that subjects #2 and #16 affect the subject-independent performance.
- Minimum classification performance in subject-dependent suggested that Boring emotion (LANV) has the lowest performance.
- Cross-validation results reported an average performance of 96.61% using the proposed method; meanwhile, the LoSo-CV classification architecture reached 98.67%.

II. MATERIALS AND METHODS

In the ER area, the EEG electrode scalp is the most popular and less invasive method to acquire the bio-electrical potential from a stimuli response. This study proposes an EEG signal processing for ER in video game playing. The input data is taken from a public dataset called GAMEEMO; this data is later passed through a denoising method based on several stages, such as a pair EEG channels selection, a regression algorithm, rotation matrices, direction vectors, and a robust feature extraction

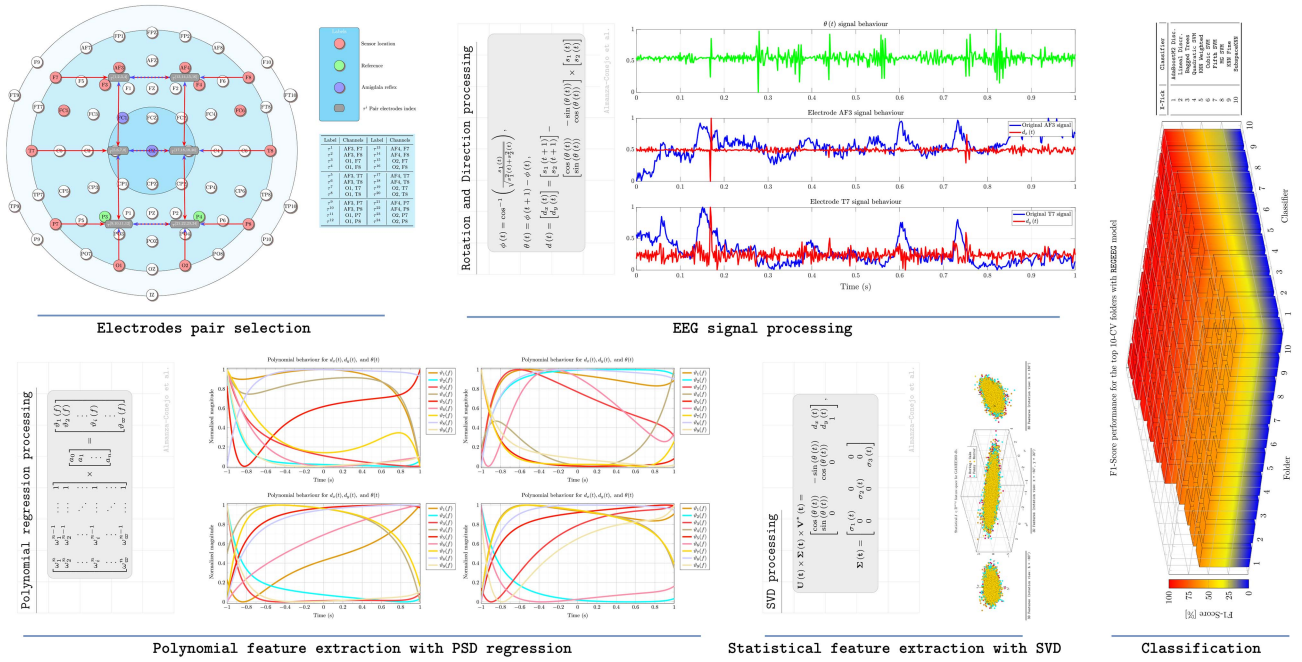


Fig. 1. General framework of the regression-based EEG signal processing method for emotion recognition.

technique providing discriminant statistical and polynomial features. Fig. 1 shows a general diagram of the proposed methodology, including noise reduction, polynomial and statistical feature extraction, and feature space behavior to provide data through 28 different ML-kernels.

A. GAMEEMO Dataset

GAMEEMO dataset developed by [24] recovered the EEG bio-electrical potential information from 28 subjects in a 20–27 age range to study ER. Each subject played four computer video game in a 5 minutes gameplay per emotion stimuli, achieving 20 gameplaying minutes per subject. Brains' activity is recorded using a 14 EMOTIV EPOC+ Fig. 2; green dots are sensor electrodes, red dots are the reference electrodes, and blue dots are the electrodes where amygdala and hippocampus information presented a significant activity electrode scalp [29]. Fig. 2 is based on a 10–20 electrode scalp distribution. Besides, the frequency rate was down-sampled to 128 Hz; later, a Butterworth filter was applied to achieve 0.16–43 Hz for noise reduction. In short, 1568 samples are recorded and stored in $\Xi(t) \in \mathbb{R}^{28 \times 4 \times 14 \times \gamma}$, where $\gamma \approx [300(s) \times 128(Hz)] \approx 38400$ samples.

B. EEG Channels Pair Structure

The proposed REEG method develops the regression signal processing per subject using a 24 EEG pair of electrodes. The core concept involves mapping an imaginary perimeter to the FC1 and CZ electrodes, where the activity of the amygdala and hippocampus is most prominently represented. Then, a signal response is estimated, considering that two mutually orthogonal EEG channels can directly influence such channels. For example, the AF3 and F7 electrodes compound the τ^1 mutually orthogonal pair electrodes. Besides, the T7 and T8 information

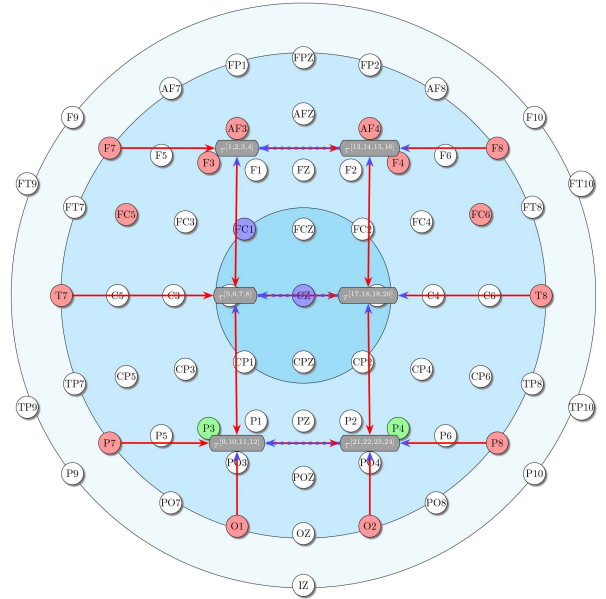


Fig. 2. Electrode scalp distribution using the EMOTIV EPOC+.

was also analyzed, mutually orthogonal to AF3 and O1 channels, to create the subset $\tau^{\{5,6,7,8\}}$. This configuration gave the best results; visually, Fig. 2 shows each pair of channel pairings. Besides, electrode pair labels are summarized in Table I.

Once paired configurations are established, each emotion stimuli is arranged in a single $\varphi_{\Xi}^{\tau_j}$ tensor array, where φ , τ , and η represent the indexing variables for subjects, emotion stimuli, pair channel selection, and EEG length, respectively. Meanwhile, j is the channel index into the i -th pair of electrodes. Hence, the main proposal of this framework is to develop an

TABLE I
ELECTRODES PAIR SELECTION LABELS FOR THE CONFIGURATIONS SHOWN IN FIG. 2

Label	Channels	Label	Channels	Label	Channels
τ^1	AF3, F7	τ^9	AF3, P7	τ^{17}	AF4, T7
τ^2	AF3, F8	τ^{10}	AF3, P8	τ^{18}	AF4, T8
τ^3	O1, F7	τ^{11}	O1, P7	τ^{19}	O2, T7
τ^4	O1, F8	τ^{12}	O1, P8	τ^{20}	O2, T8
τ^5	AF3, T7	τ^{13}	AF4, F7	τ^{21}	AF4, P7
τ^6	AF3, T8	τ^{14}	AF4, F8	τ^{22}	AF4, P8
τ^7	O1, T7	τ^{15}	O2, F7	τ^{23}	O2, P7
τ^8	O1, T8	τ^{16}	O2, F8	τ^{24}	O2, P8

outperforming REGEEG method based on a regression algorithm for noise reduction. Subsequently, the power spectrum behavior from EEGs is computed. Next, a feature extraction process is applied to each EEG input in $\varphi_{\Xi}^{\tau_j^i}$, the procedure is detailed in Section II-C.

C. REGEEG Signal Processing

A signal pre-processing and post-processing pipeline is applied to each EEG channel from the GAMEEMO dataset, represented by the tensor data $\varphi_{\Xi}^{\tau_j^i}$, using the REGEEG regression algorithm. Gameplay season is recorded for about 300 seconds following a $f_s = 128\text{Hz}$ ratio, achieving $\eta = 38252$ discrete time samples. A factor integer criterion is applied to η , yielding $\eta = \varpi \times v = 38252$. Here, $\varpi = 2 \times 2 \times 131 = 524$ corresponds to the EEG window length, and $v = 73$ indicates the number of windows. Non-overlapping window displacement is implemented during feature extraction. For a formal analysis, $s_1(t) = \tau_1^i$ and $s_2(t) = \tau_2^i$ EEG pair information is stored in a vector as

$$p(t) = \begin{bmatrix} s_1(t) \\ s_2(t) \end{bmatrix}, \quad (1)$$

then, the angular vector $\phi(t)$ is computed by

$$\phi(t) = \arccos \left\{ \frac{s_1(t)}{\sqrt{s_1^2(t) + s_2^2(t)}} \right\}, \quad (2)$$

to apply a forward difference between the samples $\phi(t)$ and $\phi(t+1)$ in discrete time as

$$\theta(t) = \Delta\phi(t) = \phi(t+1) - \phi(t), \quad (3)$$

produces the signal behavior shown in Fig. 3. Furthermore, this study proposes to use the regression described by the following rule,

$$\begin{bmatrix} p(t+1) \\ 1 \end{bmatrix} = \begin{bmatrix} R(t) & d(t) \\ \mathbf{0} & 1 \end{bmatrix} \times \begin{bmatrix} p(t) \\ 1 \end{bmatrix}, \quad (4)$$

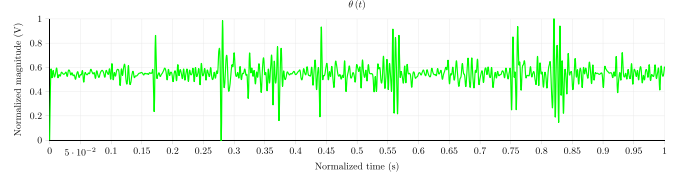
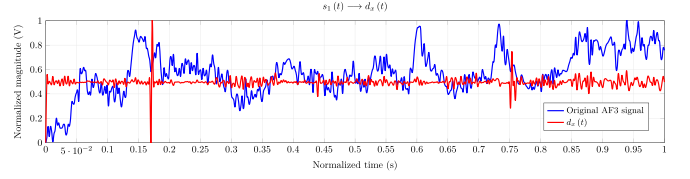
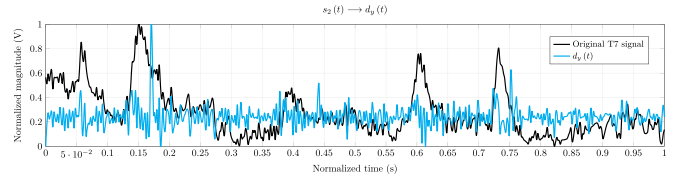


Fig. 3. $\theta(t)$ signal behavior from (3).



(a) $d_x(t)$ direction element vector.



(b) $d_y(t)$ direction element vector.

Fig. 4. Original EEG time behavior for a pair of channel electrodes and $d(t)$ denoised EEG.

where the $R(t)$ matrix rotation and $d(t)$ direction vector are computed as follows

$$R(t) = \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) \\ \sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix}, \quad (5)$$

$$d(t) = \begin{bmatrix} d_x(t) \\ d_y(t) \end{bmatrix} = p(t+1) - R(t) \times p(t). \quad (6)$$

Besides, the direction vector $d(t)$ can be explicitly rewritten as

$$d(t) = \begin{bmatrix} s_1(t+1) \\ s_2(t+1) \end{bmatrix} - \begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) \\ \sin(\theta(t)) & \cos(\theta(t)) \end{bmatrix} \times \begin{bmatrix} s_1(t) \\ s_2(t) \end{bmatrix}, \quad (7)$$

where the time-domain behavior of (7) in the ϖ window is shown in Fig. 4. Consequently, the natural logarithmic of the power spectrum (8) for the (3) and (7) signals is computed to obtain a 12-degree polynomial expansion,

$$\begin{aligned} \text{PS}(f) &= \mathcal{F}\{R_{xx}(\tau)\} \\ &= \int_{-\infty}^{\infty} \left\{ \lim_{\eta \rightarrow \infty} \frac{1}{\eta} \int_0^{\eta} x_j^*(t-\tau) x_j(t) dt \right\} e^{-i2\pi f \tau} d\tau, \end{aligned} \quad (8)$$

in this context $x^*(t-\tau) \equiv \theta^*(t-\tau)$ or $d^*(t-\tau)$, according to the use of PS for $\theta(t)$ or $d(t)$, in completing the auto-correlation function $R_{xx}(\tau)$. Then, natural logarithmic is applied through (8) to obtain $\vartheta(f) = \ln(\text{PS}(f))$, shown in Fig. 5. Finally, Vandermonde matrix is determined with the frequency vector $\omega_i = 2\pi f_i$ in (8), and the natural logarithmic PS density $\vartheta_i(f)$ vector

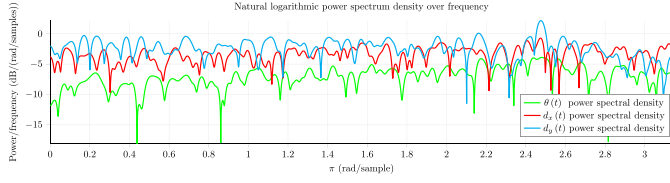


Fig. 5. Natural logarithmic of the power spectrum density per radian sample.

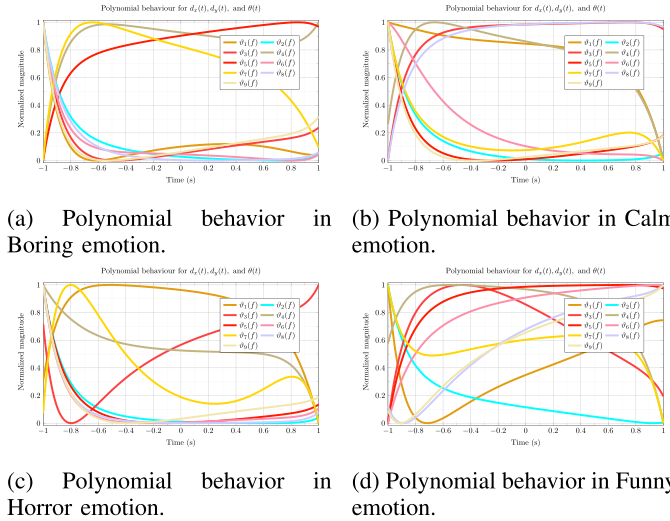


Fig. 6. Polynomial behavior of random signals per emotion stimuli in the GAMEEMO dataset.

as follows

$$\begin{bmatrix} \vartheta_1(f) \\ \vartheta_2(f) \\ \vdots \\ \vartheta_i(f) \\ \vdots \\ \vartheta_{\varpi}(f) \end{bmatrix} = \begin{bmatrix} \omega_1^n & \omega_1^{n-1} & \dots & 1 \\ \omega_2^n & \omega_2^{n-1} & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ \omega_i^n & \omega_i^{n-1} & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{\varpi}^n & \omega_{\varpi}^{n-1} & \dots & 1 \end{bmatrix} \times \begin{bmatrix} a_n \\ a_{n-1} \\ \vdots \\ a_0 \end{bmatrix}. \quad (9)$$

Therefore, the 12 polynomial degree coefficients are arranged in the vector $\mathbf{a} = [a_n \ a_{n-1} \ \dots \ a_0]^T$ by multiplying the inverse of the Vandermonde matrix and $\vartheta_i(f)$ vector. As a result, Fig. 6 shows the polynomial behavior of tree-random samples, achieving nine different patterns, using the coefficients $\mathbf{a}$ as data features per emotion stimuli in the GAMEEMO dataset. Plus, robust statistical feature extraction is computed from the eigenvalues $\Sigma(\mathbf{t})$ obtained by the Singular Value Decomposition (SVD) of the matrix (4), resulting in an augmented matrix of 3×3 composed by (5) and (6),

$$\mathbf{U} \times \Sigma \times \mathbf{V}^* = \begin{bmatrix} R(t) & d(t) \\ \mathbf{0} & 1 \end{bmatrix}. \quad (10)$$

This decomposition provides a fast and accurate representation of the diagonal eigenvalues matrix,

$$\Sigma(\mathbf{t}) = \begin{bmatrix} \sigma_1(t) & 0 & 0 \\ 0 & \sigma_2(t) & 0 \\ 0 & 0 & \sigma_3(t) \end{bmatrix}. \quad (11)$$

Notably, the eigenvalue $\sigma_2(t) = 0$ due to the SVD of (10). Besides, the feature extraction is conducted by computing discriminant statistics on the signals $\sigma_1(t)$ and $\sigma_3(t)$ as the mean, variance, median, mode, kurtosis, and skewness.

In short, a feature matrix $F \in \mathbb{R}^{\kappa \times \varrho}$ is computed, where $\kappa = 8176$ and $\varrho = 1225$ represent the total number of samples achieved in $\varphi_{\Xi}^{\tau_j^i}$ and the number of features per τ pair of electrodes, respectively. The $\varrho = \tau \times (36 + 12)$ appended polynomial coefficients and statistical features are stored as the 12-coefficients $\mathbf{ax}$ vector for the angular differential (3) and the displacement vector (7), yielding 36 polynomial coefficients. Additionally, six statistical features are also accomplished using the $\sigma_1(t)$ and $\sigma_3(t)$, resulting in 12 additional statistical features.

III. NUMERICAL RESULTS

The proposed REGEEG signal pre- and post-processing technique is a novel and robust approach for processing EEG signals, mainly applied in emotion recognition. As shown in Fig. 4, the resulting $d_x(t)$ and $d_y(t)$ displacement components presented a noise-reduced pattern, according to the pair channel's information. Then, the angular differential vector is stored in $\theta(t)$, as shown in Fig. 3. The resulting matrix $F \in \mathbb{R}^{\kappa \times \varrho}$ stored all-subjects features extracted according to four different studied emotion stimuli in the gameplay seasons. As a result, Fig. 6 gives the polynomial form per emotion in a 12 regression degree approximation; meanwhile, Fig. 7 shows the statistical features behavior for $\sigma_1(t)$ and $\sigma_3(t)$. The proposed methodology for REGEEG signal processing is available at <https://github.com/Almanza-Conejo/REGEEG> GitHub repository. Hyper-parameters and code for training-test classification are also available in the REGEEG repository.

A. Classification Results

The proposed REGEEG method was extensively tested with several classification algorithms using a PC with Intel Xeon Silver 4210R (10 digital cores, 2.4 GHz) and 208 GB RAM running in Windows 11 OS. Algorithms were coded and tuned with the ML toolbox of MATLAB V-2022b. Numerical results were export as *.txt files to design all graphics using the TikZ and PGFplots LaTeX packages.

A set of 28 ML-kernels are implemented to test the REGEEG performance as a feature extraction technique to provide an F feature matrix. Next, F feeds each tested classifier using a 30 cross-validation folder and a 90%–10% partition ratio for training and validation to reduce overfitting. So, 7259(∓ 1) and 817(± 1) samples from F are used, whereby random indices are updated before each training and validation folder per ML-kernel, respectively. According to statistical classification results, Figs. 8 and 9 suggests an accurate performance using k-NN and SVM kernels, achieving their respective performance

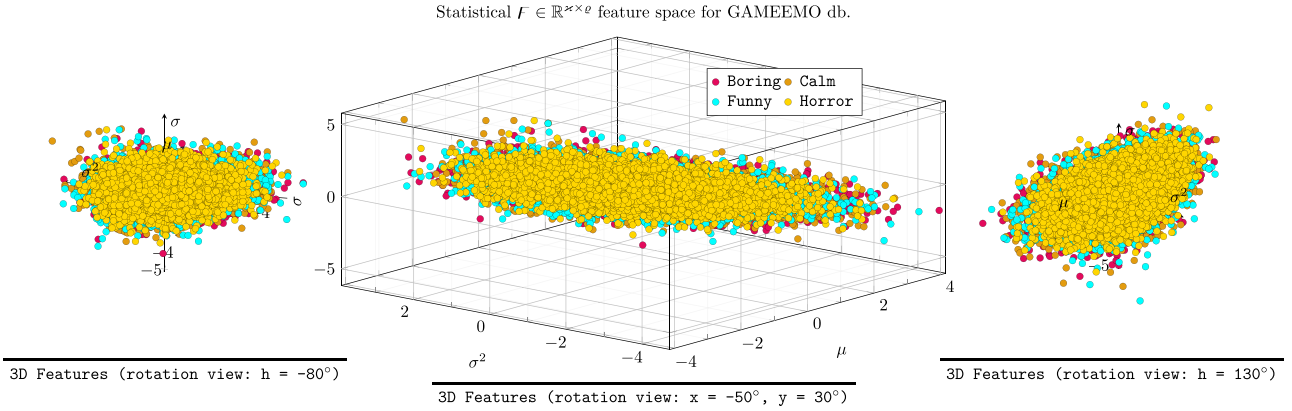
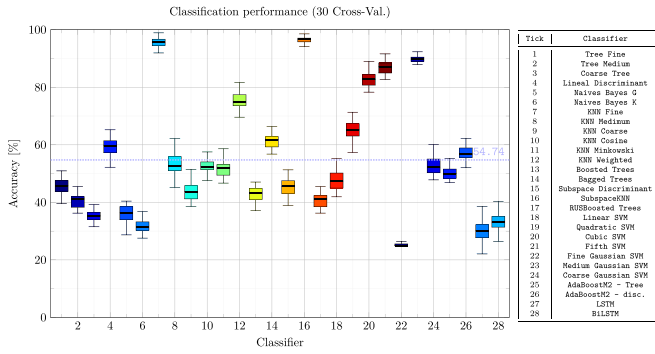
Fig. 7. Statistical feature behavior space for $\sigma_1(t)$ and $\sigma_3(t)$.

Fig. 8. Performance for 28 different kernel classification algorithms.

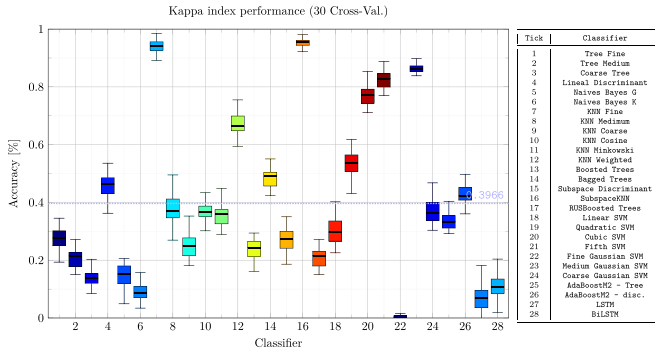


Fig. 9. Kappa-score performance for 28 different kernel classification algorithms.

metrics above the 80%, where SubspaceKNN kernel reaches the top-1 accuracy rate. Meanwhile, Fig. 10 shows the Receiver Operating Characteristic (ROC) performance for the top-1 accuracy ML-model overall iteration folders. As theory suggests, the closer the ROC curve is to the top-left corner, the more accurate the model is. Here, “Horror” achieved as the less accurate class. However, we must consider that the mean kappa-score is close to 1.00, yielding a reliable performance. In addition, the 28 ML classification kernels average is also included as a blue dotted line in Figs. 8 and 9. Additionally, we tested the validation performance for the top-10 classification kernels,

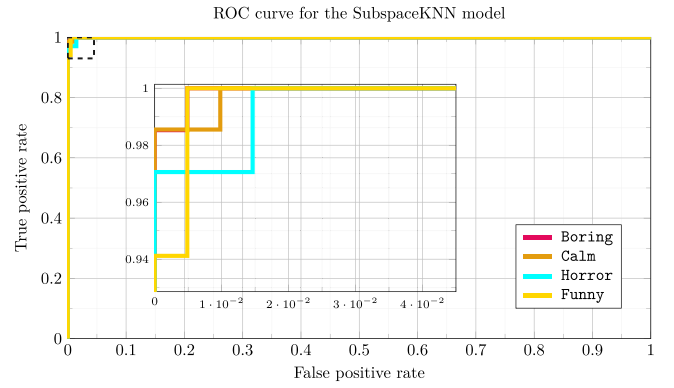
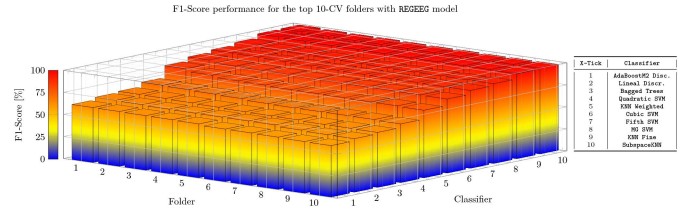


Fig. 10. ROC curve performance using the SubspaceKNN model.

Fig. 11. Top-10 classifiers across 10-CV folders using the F feature matrix.TABLE II
HYPER-PARAMETERS OF SUBSPACE-KNN MODEL

Hyper-parameter	Value
Method	Subspace
Number of Learning Cycles	30
Learner Type	kNN
Number of Predictors to Sample	$\max(1, \min(108, \text{size}(\text{predictors}, 2) - 1))$
Class Names	classNames

presented by ascend-sorting in Fig. 11, across 10-CV folders. SubspaceKNN is an ensemble model of a specific number of trained kernels into a subspace model. Here, each of these kernels takes a subset of feature samples to reduce overfitting, ensuring diversity of samples in the F matrix. For reproducible results, we included the SubspaceKNN hyper-parameters in Table II and the rest of the top-5 hyper-parameters kernels in

TABLE III
MIN-MAX CONFUSION MATRIX (CM) AND CLASSIFICATION PERFORMANCE PER ML K-NN KERNEL MODEL

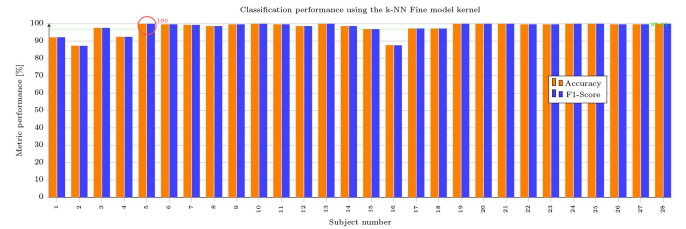
k-NN Fine Performance $\left(\mu_t = 0.1468, \sigma_t^2 = 0.0140\right)$											
$\min_{i=2}^f \alpha(i)$ classification performance CM						$\max_{i=5}^f \alpha(i)$ classification performance CM					
	Boring	Calm	Horror	Funny	Total	Boring	Calm	Horror	Funny	Total	
Boring	70	1	2	0	73	73	0	0	0	73	
Calm	7	57	4	5	73	0	73	0	0	73	
Horror	6	0	67	0	73	0	0	73	0	73	
Funny	3	5	4	61	73	0	0	0	73	73	
Evaluation metrics [%]					$\mu \pm \sigma$	Evaluation metrics [%]					$\mu \pm \sigma$
Precision	81.40	90.48	87.01	92.42	87.83 $\pm$ 04.84	100.00	100.00	100.00	100.00	100.00 $\mp$ 0.00	
Recall	95.89	78.08	91.78	83.56	87.33 $\pm$ 08.02	100.00	100.00	100.00	100.00	100.00 $\mp$ 0.00	
F1-Score	88.05	83.82	89.33	87.77	87.24 $\pm$ 02.38	100.00	100.00	100.00	100.00	100.00 $\mp$ 0.00	
Accuracy	87.33					100.00					
k-NN Subspace Performance $\left(\mu_t = 0.1330, \sigma_t^2 = 0.0129\right)$											
$\min_{i=2}^s \alpha(i)$ classification performance CM						$\max_{i=5}^s \alpha(i)$ classification performance CM					
	Boring	Calm	Horror	Funny	Total	Boring	Calm	Horror	Funny	Total	
Boring	72	0	1	0	73	73	0	0	0	73	
Calm	8	64	0	1	73	0	73	0	0	73	
Horror	1	0	72	0	73	0	0	73	0	73	
Funny	5	9	2	57	73	0	0	0	73	73	
Evaluation metrics [%]					$\mu \pm \sigma$	Evaluation metrics [%]					$\mu \pm \sigma$
Precision	83.72	87.67	96.00	98.28	88.17 $\pm$ 05.84	100.00	100.00	100.00	100.00	100.00 $\mp$ 0.00	
Recall	98.63	87.67	98.63	78.08	87.67 $\pm$ 08.66	100.00	100.00	100.00	100.00	100.00 $\mp$ 0.00	
F1-Score	90.57	87.67	97.30	87.02	87.57 $\pm$ 03.99	100.00	100.00	100.00	100.00	100.00 $\mp$ 0.00	
Accuracy	90.75					100.00					

the https://github.com/Almanza-Conejo/REGEEG/blob/main/code/REGEEG_classifiers.m GitHub repository due to paper space constraints.

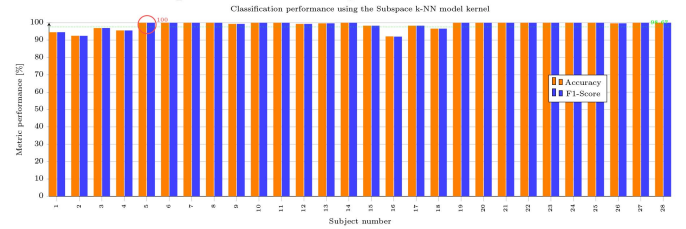
B. Leave One Subject Out Cross Validation

In close connection, a Leave One Subject Out Cross Validation (LoSo-CV) is performed to evaluate the relevance classification performance per subject. The LoSo-CV used data features per subject as a validation subset, leaving the rest of the subjects' features for training. According to boxplot Fig. 8, Fine k-NN and SubspacekNN classification models performed the best. Both outperforming classification kernels were used to test the LoSo-CV REGEEG approach, yielding higher reliability to this approach based on the accuracy and kappa-score performances per trained model, as shown in Fig. 9. Then, Table III includes the min-max training performance, Confusion Matrix (CM), and mean and variance time performances. Meanwhile, accuracy and F1-score are displayed in Fig. 12(a) and (b).

Both bar charts in Fig. 12 presents almost the same performance per subject-out of the training using two different kNN kernels, achieving the $\max_{i \in [a,b]} \alpha(i) = 100$ in both kernels. Here, α is the i -th subject-out accuracy and $[a, b]$ is the interval between the first and the last subject subset performance. In particular, the $\min_{i \in [a,b]} \alpha(i) \approx 90$ for both kernels. However, SubspaceKNN exceeded the metric, and Fine k-NN obtained



(a) LoSo-CV performance using the k-NN fine kernel model.



(b) LoSo-CV performance using the Subspace k-NN kernel model.

Fig. 12. LoSo classification performance per k-NN kernel model.

inferior results. Hence, $\mu_{\alpha_f(i)} = 97.91\%$ and $\mu_{\alpha_s(i)} = 98.67\%$ are the average subject-dependent accuracy performances. Additionally, the difference $\delta_s(\max_{i=5}^s, \min_{i=2}^s) = 9.25\%$ was estimated for the subject-dependent classification rate for SubspaceKNN model. Then, $\delta_f(\max_{i=5}^f, \min_{i=2}^f) = 12.67\%$ was

TABLE IV
PERFORMANCE COMPARISON AMONG REPRESENTATIVE ER APPROACHES

ER works in the state-of-the-art.						
Reference	Dataset	AI-Approach	EC	Classes	Accuracy [%]	F1 [%]
Joshi et al. [11]	SEED	BiLSTM	62	PNN	76.29	–
Zheng et al. [13]	SEED	SVM	62	PNN	85.23	–
Asadzadeh et al. [12]	SEED	DGCNN	62	PNN	98.51	–
Aslan et al. [25]	GAMEEMO	CNN	14	BCHF	95.30	–
Aslan et al. [27]	GAMEEMO	SVM	14	BCHF	98.78	98.78
Tuncer et al. [26]	GAMEEMO	SVM	1	BCHF	99.82	99.91
REGEEG method	GAMEEMO	k-NN Subspace	10	BCHF	96.61	96.61

achieved for the subject-dependent kNN fine model (less is better).

IV. DISCUSSION

REGEEG is a robust signal processing method for analyzing EEGs. Noise artifacts are reduced to obtain a 12 polynomial degree representation and a statistical feature set. Initially, the importance of mapping the amygdala area should be proven. According to [29], the FC1 and CZ electrodes are the relevant EEG channels that reflect the energy of the amygdala and hippocampus. In this order of ideas, the pair of electrode selection is consistent with the energy influence of two mutually orthogonal electrodes.

Even if the energy propagation of a neuron is not linear, a linear propagation can be approximated, where the vertical and horizontal intersection forms an orthogonal behavior between two electrodes. For example, it can be established that the energy propagation of AF3 reaches the O1 electrode (red arrow of Fig. 2). The AF3 and O1 signal propagation is intercepted by the F7, F8, T7, T8, P7, and P8 signals. This process generates four pairs of electrodes per interception point. Once the pair electrodes are set up, the REGEEG is implemented per v window to compute the angular differential vector (3) and the displacement vector (7), where a noise reduction effect is notable in Fig. 4. These two signals compute the regression coefficients to describe the (9) polynomial behavior per emotion in GAMEEMO dataset. Finally, statistical features are calculated by following the criteria in (10). Both, polynomial and statistical features, were computed using (5) and (7).

Average performance per classification kernel using the F matrix suggested an extremely discriminant feature space. According to Fig. 8, SVM-based algorithms presented an increasing performance proportional to the kernel's degree augmentation; Cubic, Fifth, and Medium Gaussian kernels reported an acceptable performance, although Medium Gaussian included several outliers. The cross-validation folders presented a 96.6122% average performance for the four classes. Here, 30 different folders are tested with a 98.8971% maximum performance.

In this order of ideas, a LoSo-CV is performed to test the performance per subject, resulting in subjects #2 and #16 (in the k-NN Fine model) as the lower performance. This performance is equally presented in Subspace k-NN kernel. The achieved performance for both subjects could be attributed to a lower interest in gameplay due to a certain unfamiliarity with the game,

similar computer video games, or any other physical- or social-conflict interest.

In short, most of the subjects produced about 100% accuracy and F1 score, suggesting suitable data acquisition understanding and well-being conditions. Table IV briefly compares other ER techniques, where [26] presents an accurate ER method using the GAMEEMO dataset.

A. Principal Findings

The contribution between statistical features and polynomial coefficients describes a satisfactory performance using the τ^i pair channels selection method, implemented initially to map a closely related area to the amygdala and hippocampus EEG channels (FCZ and CZ). Here, an orthogonal signal behavior is emphasized between one and another selected channels. In this way, we could achieve the pair channel's intersection and their influence in a specific brain area, see Fig. 2.

In this respect, the relation between the spatial signal behavior and a numerical description based on a regression algorithm is performed by the REGEEG approach. This algorithm performs a noise reduction and an input pattern modification to the original EEG signals (Fig. 4). Close related, (3) describes the potential difference flux between both channels into the τ^i . Then, the statistical features and polynomial coefficients are computed following (7) and (10). Posteriorly, F feeds several ML classification kernels, whereby LoSo-CV found that subject #2 and #16 features subset reduces the performance into a subject-independent classification rate. In addition, the training and validation time performance is presented in Table III, where average time suggests a speedier training using the k-NN Subspace kernel, concluding that the subspace kernel is computationally the most accurate and faster for this proposal.

B. Advantages

The EEG signal length in the GAMEEMO dataset is constant (≈ 300 seconds). This choice yields constant window analysis per EEG for the REGEEG algorithm, carrying out the same number of samples per emotion. The noise reduction based on the proposed $d(t)$ signal behavior suggested a different implementation in contrast to EMD or WD approaches; nevertheless, an efficient and robust comparison between Fourier, wavelet, and REGEEG is necessary. The F feature matrix distribution, presented as a long column-element vector, ϱ size, is also

remarkable. For this study, a $\kappa = 196224$ and $\varrho = 51$ matrix distribution was initially tested, achieving an inaccurate performance (lower than 40% for all proposed kernels). Instead, we detected the correlation between features and pair channel selection. Here, based on the main channel selection approach, we inferred the importance of using each electrode pair feature as a sample of emotions and subjects (considering the v window). This inference is supported by the excellent numerical results obtained using the proposed REGEEG method. Such a proposal produced an accurate emotion recognition model, testing the pair channel selection method, and supported the main idea that mapping the FC1 and CZ electrodes with a linear approximation energy's propagation can influence rounded electrodes. We also proposed a top-3 ML-kernels that outperforms the REGEEG methodology; however, testing the proposed regression methodology and feature extraction in different datasets must be deeply studied.

C. Limitations

In this study, the electrode pair notation suggests an orthogonal intersection between two EEG channels, but the main disadvantage is that the brain signals are not linear. It was systematically performed this algorithm based on a linear EEG energy distribution, but newer perspectives in energy propagation based on brain connections are necessary. Plus, the processing time for feature extraction and training tasks is complex. The computed regression per pair electrodes for each window processing becomes REGEEG an exhaustive algorithm. For this reason, a feature selection algorithm can be tested, such as Minimum Redundancy, Maximum Relevance, or Relief, to explore the significance of each pair channel electrodes here proposed. This strategy can improve the REGEEG signal processing and feature extraction time performance, reducing feature outliers. This last task can benefit the classification, even making that classification kernels improve their performance. Nevertheless, this suggestion is part of another study.

V. CONCLUSION

REGEEG is an EEG signal processing approach with the leading proposal to map a closer area to the amygdala and hippocampus. This method supplies an understanding of filtering and feature extraction based on statistical and polynomial coefficients. Furthermore, the filtering technique where $\theta(t)$ and $d(t)$ are processed, (3) and (7), showed less sensitive high-frequency changing patterns and a closely related dB magnitude in the power spectrum density in the frequency domain (Fig. 5).

The classification results suggest a high accuracy rate in a 30-folder cross-validation technique. Meanwhile, the LoSo-CV explains that subjects #2 and #16 are considerable outliers in the signal acquisition experiment. This effect could be attributed to subjects' non-optimal psychological or physical behavior.

However, a subject-dependent classification technique achieves a 98.67% accuracy, close to the subject-independent performance (96.61%). However, the proposed feature extraction technique showed excellent classification performance in

5/28 of the classifiers' kernels, suggesting a deep analysis in another dataset or feature reduction methods. Findings collectively highlight the impact of the pair channels array on developing an emotion recognition model, emphasizing the importance of mapping the amygdala's perimeter for precise statistical and polynomial feature extraction and model's reliability.

AUTHOR CONTRIBUTIONS

Oscar Almanza-Conejo and Mario Alberto Ibarra-Manzano conceptualized the study, designed the methodology, developed the algorithms, conducted the experiments, performed data analysis, and wrote the original manuscript. Juan Gabriel Avina-Cervantes and Arturo Garcia-Perez provided feedback, contributed to the revision of the manuscript, and supported the refinement of the research framework. All authors reviewed and approved the final version of the manuscript.

ACKNOWLEDGMENT

The authors are grateful to the Engineering Division of the Campus Irapuato-Salamanca (DICIS) and the Department of Electronics Engineering from the University of Guanajuato. A special mention to the Secretaría de Ciencia, Humanidades, Tecnología e Innovación (Secihti) by the doctoral grant #815485. Authors thank the POA Electronics Department, University of Guanajuato, 2025.

REFERENCES

- [1] P. Ekman and H. Oster, "Facial expressions of emotion," *Annu. Rev. Psychol.*, vol. 30, no. 1, pp. 527–554, Jan. 1979.
- [2] E. H. Houssein, A. Hammad, and A. A. Ali, "Human emotion recognition from EEG-based braincomputer interface using machine learning: A comprehensive review," *Neural Comput. Appl.*, vol. 34, no. 15, pp. 12527–12557, May 2022.
- [3] M.-P. Hosseini, A. Hosseini, and K. Ahi, "A review on machine learning for EEG signal processing in bioengineering," *IEEE Rev. Biomed. Eng.*, vol. 14, pp. 204–218, 2021.
- [4] M. Karnati, A. Seal, D. Bhattacharjee, A. Yazidi, and O. Krejcar, "Understanding deep learning techniques for recognition of human emotions using facial expressions: A comprehensive survey," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–31, 2023.
- [5] W. Zehra, A. R. Javed, Z. Jalil, H. U. Khan, and T. R. Gadekallu, "Cross corpus multi-lingual speech emotion recognition using ensemble learning," *Complex Intell. Syst.*, vol. 7, no. 4, pp. 1845–1854, Jan. 2021.
- [6] B. J. Abbaschian, D. Sierra-Sosa, and A. Elmaghraby, "Deep learning techniques for speech emotion recognition, from databases to models," *Sensors*, vol. 21, no. 4, Feb. 2021, Art. no. 1249.
- [7] A. Topic and M. Russo, "Emotion recognition based on EEG feature maps through deep learning network," *Eng. Sci. Technol., Int. J.*, vol. 24, no. 6, pp. 1442–1454, Dec. 2021.
- [8] Y. Fang, H. Yang, X. Zhang, H. Liu, and B. Tao, "Multi-feature input deep forest for EEG-based emotion recognition," *Front. Neurobot.*, vol. 14, pp. 1–11, Jan. 2021.
- [9] P. V. and A. Bhattacharyya, "Human emotion recognition based on timefrequency analysis of multivariate EEG signal," *Knowl.-Based Syst.*, vol. 238, Feb. 2022, Art. no. 107867.
- [10] N. Zhuang, Y. Zeng, L. Tong, C. Zhang, H. Zhang, and B. Yan, "Emotion recognition from EEG signals using multidimensional information in EMD domain," *BioMed Res. Int.*, pp. 1–9, Aug. 2017.
- [11] V. M. Joshi and R. B. Ghongade, "IDEA: Intellect database for emotion analysis using EEG signal," *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 34, no. 7, pp. 4433–4447, Jul. 2022.
- [12] S. Asadzadeh, T. Y. Rezaii, S. Beheshti, and S. Meshgini, "Accurate emotion recognition using Bayesian model based EEG sources as dynamic graph convolutional neural network nodes," *Sci. Rep.*, vol. 12, no. 1, Jun. 2022, Art. no. 10282.

- [13] W. Zheng, "Multichannel EEG-based emotion recognition via group sparse canonical correlation analysis," *IEEE Trans. Cogn. Devel. Syst.*, vol. 9, no. 3, pp. 281–290, Sep. 2017.
- [14] X. Zheng, X. Liu, Y. Zhang, L. Cui, and X. Yu, "A portable HCI system-oriented EEG feature extraction and channel selection for emotion recognition," *Int. J. Intell. Syst.*, vol. 36, no. 1, pp. 152–176, Oct. 2020.
- [15] O. Almanza-Conejo, J. G. Avina-Cervantes, A. Garcia-Perez, and M. A. Ibarra-Manzano, "A channel selection method to find the role of the amygdala in emotion recognition avoiding conflict learning in EEG signals," *Eng. Appl. Artif. Intell.*, vol. 126, Nov. 2023, Art. no. 106971.
- [16] X. Wu, W.-L. Zheng, Z. Li, and B.-L. Lu, "Investigating EEG-based functional connectivity patterns for multimodal emotion recognition," *J. Neural Eng.*, vol. 19, no. 1, Jan. 2022, Art. no. 016012.
- [17] G. Zhang, V. Davoodnia, and A. Etemad, "PARSE: Pairwise alignment of representations in semi-supervised EEG learning for emotion recognition," *IEEE Trans. Affect. Comput.*, vol. 13, no. 4, pp. 2185–2200, Oct./Dec. 2022.
- [18] N. K. Al-Qazzaz, M. K. Sabir, S. H. B. M. Ali, S. A. Ahmad, and K. Grammer, "Electroencephalogram profiles for emotion identification over the brain regions using spectral, entropy and temporal biomarkers," *Sensors*, vol. 20, no. 1, Dec. 2019, Art. no. 59.
- [19] A. Khosla, P. Khandnor, and T. Chand, "A comparative analysis of signal processing and classification methods for different applications based on EEG signals," *Biocybernetics Biomed. Eng.*, vol. 40, no. 2, pp. 649–690, Apr. 2020.
- [20] A. Sakalle, P. Tomar, H. Bhardwaj, D. Acharya, and A. Bhardwaj, "A LSTM based deep learning network for recognizing emotions using wireless brainwave driven system," *Expert Syst. with Appl.*, vol. 173, Jul. 2021, Art. no. 114516.
- [21] Y. Xie and S. Oniga, "A review of processing methods and classification algorithm for EEG signal," *Carpathian J. Electron. Comput. Eng.*, vol. 13, no. 1, pp. 23–29, Sep. 2020.
- [22] H. He, Y. Tan, J. Ying, and W. Zhang, "Strengthen EEG-based emotion recognition using firefly integrated optimization algorithm," *Appl. Soft Comput.*, vol. 94, Sep. 2020, Art. no. 106426.
- [23] E. P. Torres, E. A. Torres, M. Hernández-Álvarez, and S. G. Yoo, "EEG-based BCI emotion recognition: A survey," *Sensors*, vol. 20, no. 18, Sep. 2020, Art. no. 5083.
- [24] T. B. Alakus, M. Gonen, and I. Turkoglu, "Database for an emotion recognition system based on EEG signals and various computer games GAMEEMO," *Biomed. Signal Process. Control*, vol. 60, Jul. 2020, Art. no. 101951.
- [25] M. Aslan, M. Baykara, and T. B. Alakus, "Analysis of brain areas in emotion recognition from EEG signals with deep learning methods," *Multimedia Tools Appl.*, vol. 83, Sep. 2023, pp. 32423–32452.
- [26] T. Tuncer, S. Dogan, and A. Subasi, "A new fractal pattern feature generation function based emotion recognition method using EEG," *Chaos, Solitons Fractals*, vol. 144, Mar. 2021, Art. no. 110671.
- [27] M. Aslan, "CNN based efficient approach for emotion recognition," *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 34, no. 9, pp. 7335–7346, Oct. 2022.
- [28] O. Almanza-Conejo, D. L. Almanza-Ojeda, J. L. Contreras-Hernandez, and M. A. Ibarra-Manzano, "Emotion recognition in EEG signals using the continuous wavelet transform and CNNs," *Neural Comput. Appl.*, vol. 35, no. 2, pp. 1409–1422, Sep. 2022.
- [29] J. Albert, S. López-Martín, and L. Carretié, "Emotional context modulates response inhibition: Neural and behavioral data," *NeuroImage*, vol. 49, no. 1, pp. 914–921, Jan. 2010.